Reg. No. | | | | | | | | | | | [13EC63/12EC63]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

ANTENNAS AND WAVE PROPAGATION

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any FIVE full questions

1.	**a.**	Define the following terms: **i.** Radiation intensity **ii.** Beam efficiency **iii.** Directivity **iv.** Effective height	**08**
	b.	An antenna has a field pattern given by $E(\theta)=\cos^2\theta$ for $0^0 \leq \theta \leq 90^o$. Find **i.** The half power beam width **ii.** Beam area	**06**
2.	**a.**	Derive the expression for the field of an array of two isotropic point sources of equal amplitude with phase quadrature. Assume the spacing between the point sources be $\lambda/2$.	**07**
	b.	What do you mean by 'Hansen Wood yard condition'? Prove that the first null for a broadside array occurs at $\pm\lambda/nd$.	**07**
3.	**a.**	Explain the principle of pattern multiplication with an example.	**05**
	b.	Prove that the radiation resistance of a $\lambda/2$ dipole is 73Ω.	**09**
4.	**a.**	Derive the expression for far field pattern of a small loop antenna.	**08**
	b.	Prove that the radiation resistance of a small loop having one or more turns is given by $R_r = 31200\left(\frac{nA}{\lambda^2}\right)^2 \Omega$	**06**
5.	**a.**	Write a brief note on patch antenna.	**05**
	b.	Explain about the practical considerations for design of monofilar Axial mode Helical antenna.	**06**
	c.	A helix has tubing width of 5mm. Find the required thickness of a polystyrene sheet for matching to a 50Ω co-axial transmission line Take $\varepsilon_r=2.7$ for polystyrene sheet.	**03**
6.	**a.**	How the feeding arrangements are done using $\lambda/2$ dipole antenna in Turnstile antennas? Explain with neat figures & relevant expressions.	**07**
	b.	Write a note on Ultra Wide Band (UWB) antennas.	**07**
7.	**a.**	What are the different modes of wave propagations? Explain with neat diagrams.	**06**
	b.	Prove that the refractive index of ionosphere is less than 1 and is inversely proportional to iconic density.	**08**
8.	**a.**	A VHF communications is to be established with 35W transmitter at 90MHZ. Find the distance upto which line of sight distance communication may be possible if the heights of transmitting & receiving antennas are 40m & 25m respectively also determine the field strength at the receiver.	**06**
	b.	Write short notes on: **i.** Tropospheric scatter propagation. **ii.** Critical frequency.	**08**